Conic Sections

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Degrees, radians, and angles as rotations**

A geometric figure scaled by factor $k = 2$ has an original side length of 4 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 8 cm. Multiplying the original length by the scale factor gives $4 \times 2 = 8$ cm.

2 **[Foundational]** • **Target: Degrees, radians, and angles as rotations**

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

Final Answer: 14. Substituting $x = 1$ into $y = 5x + 9$ gives $y = 5(1) + 9 = 14$.

3 **[Foundational]** • **Target: Degrees, radians, and angles as rotations**

In ΔABC studying degrees, radians, and angles as rotations, $m\angle A = 35^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 93°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (35^\circ + 52^\circ) = 93^\circ$.

4 **[Foundational]** • **Target: Degrees, radians, and angles as rotations**

A geometric figure scaled by factor $k = 3$ has an original side length of 4 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 12 cm. Multiplying the original length by the scale factor gives $4 \times 3 = 12$ cm.

5 **[Foundational]** • **Target: Degrees, radians, and angles as rotations**

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

Final Answer: 19. Substituting $x = 2$ into $y = 5x + 9$ gives $y = 5(2) + 9 = 19$.

6 **[Foundational]** • **Target: Degrees, radians, and angles as rotations**

In ΔABC studying degrees, radians, and angles as rotations, $m\angle A = 35^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 86°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (35^\circ + 59^\circ) = 86^\circ$.

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7 [Foundational] • Target: Degrees, radians, and angles as rotations

A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 16 cm. Multiplying the original length by the scale factor gives $4 \times 4 = 16$ cm.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Degrees, radians, and angles as rotations

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.

Final Answer: 24. Substituting $x = 3$ into $y = 5x + 9$ gives $y = 5(3) + 9 = 24$.

9 [Intermediate] • Target: Degrees, radians, and angles as rotations

In ΔABC studying degrees, radians, and angles as rotations, $m\angle A = 35^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 79°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (35^\circ + 66^\circ) = 79^\circ$.

10 [Intermediate] • Target: Degrees, radians, and angles as rotations

A geometric figure scaled by factor $k = 5$ has an original side length of 4 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 20 cm. Multiplying the original length by the scale factor gives $4 \times 5 = 20$ cm.

11 [Intermediate] • Target: Degrees, radians, and angles as rotations

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 4$? Show your complete algebraic work.

Final Answer: 29. Substituting $x = 4$ into $y = 5x + 9$ gives $y = 5(4) + 9 = 29$.

12 [Intermediate] • Target: Trigonometric functions in the coordinate plane

For the model $f(x) = 3x + 3$ in trigonometric functions in the coordinate plane, evaluate $f(2)$.

Final Answer: 9. Substitute $x = 2$: $f(2) = 3(2) + 3 = 9$.

13 [Intermediate] • Target: Trigonometric functions in the coordinate plane

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

Final Answer: $x = 1$ and $x = 3$. Factoring $(x - 1)(x - 3) = 0$ yields the roots $x = 1$ and $x = 3$.

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14 [Intermediate] • Target: Trigonometric functions in the coordinate plane

A data set recorded for trigonometric functions in the coordinate plane consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 18.5. The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Trigonometric functions in the coordinate plane • Cross-Lesson: 2.01 Quadratics & Complex Numbers

For the model $f(x) = 3x + 3$ in trigonometric functions in the coordinate plane, evaluate $f(3)$.

Final Answer: 12. Substitute $x = 3$: $f(3) = 3(3) + 3 = 12$.

16 [Challenging] • Target: Trigonometric functions in the coordinate plane • Cross-Lesson: 3.01 Polynomial Functions

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

Final Answer: $x = 2$ and $x = 4$. Factoring $(x - 2)(x - 4) = 0$ yields the roots $x = 2$ and $x = 4$.

17 [Challenging] • Target: Trigonometric functions in the coordinate plane • Cross-Lesson: 4.01 Rational Expressions

For the model $f(x) = 3x + 3$ in trigonometric functions in the coordinate plane, evaluate $f(4)$.

Final Answer: 15. Substitute $x = 4$: $f(4) = 3(4) + 3 = 15$.

18 [Challenging] • Target: Trigonometric functions in the coordinate plane • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

Final Answer: $x = 3$ and $x = 5$. Factoring $(x - 3)(x - 5) = 0$ yields the roots $x = 3$ and $x = 5$.

19 [Challenging] • Target: Trigonometric functions in the coordinate plane • Cross-Lesson: 6.01 Radicals & Rational Exponents

For the model $f(x) = 3x + 3$ in trigonometric functions in the coordinate plane, evaluate $f(5)$.

Final Answer: 18. Substitute $x = 5$: $f(5) = 3(5) + 3 = 18$.

20 [Challenging] • Target: Trigonometric functions in the coordinate plane • Cross-Lesson: 7.01 Sequences & Series

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

Final Answer: $x = 4$ and $x = 6$. Factoring $(x - 4)(x - 6) = 0$ yields the roots $x = 4$ and $x = 6$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Trigonometric functions as right triangle ratios

In ΔABC studying trigonometric functions as right triangle ratios, $m\angle A = 33^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 102° ✓

b. 112°

c. 94°

d. 147°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (33^\circ + 45^\circ) = 102^\circ$.

22 [Foundational] • Target: Trigonometric functions as right triangle ratios

A geometric figure scaled by factor $k = 3$ has an original side length of 7 cm. What is the corresponding image length?

a. 18 cm

b. 21 cm ✓

c. 28 cm

d. 63 cm

Multiplying the original length by the scale factor gives $7 \times 3 = 21$ cm.

23 [Foundational] • Target: Trigonometric functions as right triangle ratios

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 7)$. What is its y-coordinate when $x = 3$?

a. 13

b. 18

c. 16 ✓

d. 19

Substituting $x = 3$ into $y = 3x + 7$ gives $y = 3(3) + 7 = 16$.

24 [Foundational] • Target: Trigonometric functions as right triangle ratios

In ΔABC studying trigonometric functions as right triangle ratios, $m\angle A = 33^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 81° ✓

b. 91°

c. 73°

d. 147°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (33^\circ + 66^\circ) = 81^\circ$.

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25 [Foundational] • Target: Special right triangles and exact value trigonometric ratios

In ΔABC studying special right triangles and exact value trigonometric ratios, $m\angle A = 34^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 101° ✓

b. 111°

c. 93°

d. 146°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (34^\circ + 45^\circ) = 101^\circ$.

26 [Foundational] • Target: Special right triangles and exact value trigonometric ratios

A geometric figure scaled by factor $k = 3$ has an original side length of 8 cm. What is the corresponding image length?

a. 21 cm

b. 24 cm ✓

c. 32 cm

d. 72 cm

Multiplying the original length by the scale factor gives $8 \times 3 = 24$ cm.

27 [Foundational] • Target: Special right triangles and exact value trigonometric ratios

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 3$?

a. 16

b. 22

c. 20 ✓

d. 24

Substituting $x = 3$ into $y = 4x + 8$ gives $y = 4(3) + 8 = 20$.

28 [Foundational] • Target: Special right triangles and exact value trigonometric ratios

In ΔABC studying special right triangles and exact value trigonometric ratios, $m\angle A = 34^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 80° ✓

b. 90°

c. 72°

d. 146°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (34^\circ + 66^\circ) = 80^\circ$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Trigonometric functions as right triangle ratios

A geometric figure scaled by factor $k = 2$ has an original side length of 7 cm. What is the corresponding image length?

a. 12 cm

b. 14 cm ✓

c. 21 cm

d. 28 cm

Multiplying the original length by the scale factor gives $7 \times 2 = 14$ cm.

30 [Intermediate] • Target: Trigonometric functions as right triangle ratios

In ΔABC studying trigonometric functions as right triangle ratios, $m\angle A = 33^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 95° ✓

b. 105°

c. 87°

d. 147°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (33^\circ + 52^\circ) = 95^\circ$.

31 [Intermediate] • Target: Trigonometric functions as right triangle ratios

In ΔABC studying trigonometric functions as right triangle ratios, $m\angle A = 33^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 88° ✓

b. 98°

c. 80°

d. 147°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (33^\circ + 59^\circ) = 88^\circ$.

32 [Intermediate] • Target: Trigonometric functions as right triangle ratios

A geometric figure scaled by factor $k = 5$ has an original side length of 7 cm. What is the corresponding image length?

a. 30 cm

b. 35 cm ✓

c. 42 cm

d. 175 cm

Multiplying the original length by the scale factor gives $7 \times 5 = 35$ cm.

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33 [Intermediate] • Target: Special right triangles and exact value trigonometric ratios

A geometric figure scaled by factor $k = 2$ has an original side length of 8 cm. What is the corresponding image length?

a. 14 cm

b. 16 cm ✓

c. 24 cm

d. 32 cm

Multiplying the original length by the scale factor gives $8 \times 2 = 16$ cm.

34 [Intermediate] • Target: Special right triangles and exact value trigonometric ratios

In ΔABC studying special right triangles and exact value trigonometric ratios, $m\angle A = 34^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 94° ✓

b. 104°

c. 86°

d. 146°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (34^\circ + 52^\circ) = 94^\circ$.

35 [Intermediate] • Target: Special right triangles and exact value trigonometric ratios

A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length?

a. 28 cm

b. 32 cm ✓

c. 40 cm

d. 128 cm

Multiplying the original length by the scale factor gives $8 \times 4 = 32$ cm.

36 [Intermediate] • Target: Special right triangles and exact value trigonometric ratios

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 4$?

a. 20

b. 26

c. 24 ✓

d. 28

Substituting $x = 4$ into $y = 4x + 8$ gives $y = 4(4) + 8 = 24$.

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37 [Intermediate] • Target: Degrees, radians, and angles as rotations

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 1$?

a. 9

b. 16

c. 14 ✓

d. 19

Substituting $x = 1$ into $y = 5x + 9$ gives $y = 5(1) + 9 = 14$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Degrees, radians, and angles as rotations • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 2$?

a. 14

b. 21

c. 19 ✓

d. 24

Substituting $x = 2$ into $y = 5x + 9$ gives $y = 5(2) + 9 = 19$.

39 [Challenging] • Target: Degrees, radians, and angles as rotations • Cross-Lesson: 6.01 Radicals & Rational Exponents

A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length?

a. 12 cm

b. 16 cm ✓

c. 20 cm

d. 64 cm

Multiplying the original length by the scale factor gives $4 \times 4 = 16$ cm.

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40 [Challenging] • Target: Degrees, radians, and angles as rotations • Cross-Lesson: 7.01 Sequences & Series

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 4$?

a. 24

b. 31

c. 29 ✓

d. 34

Substituting $x = 4$ into $y = 5x + 9$ gives $y = 5(4) + 9 = 29$.

41 [Challenging] • Target: Trigonometric functions on the unit circle • Cross-Lesson: 9.01 Trigonometric Functions

A data set recorded for trigonometric functions on the unit circle consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

42 [Challenging] • Target: Trigonometric functions on the unit circle • Cross-Lesson: 1.01 Functions & Their Graphs

For the model $f(x) = 4x + 4$ in trigonometric functions on the unit circle, evaluate $f(4)$.

a. 20 ✓

b. 24

c. 16

d. 24

Substitute $x = 4$: $f(4) = 4(4) + 4 = 20$.

43 [Challenging] • Target: Trigonometric functions on the unit circle • Cross-Lesson: 10.01 Matrices & Systems

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ **and** $x = 7$ ✓

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

Factoring $(x - 4)(x - 7) = 0$ yields the roots $x = 4$ and $x = 7$.

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44 [Challenging] • Target: Evaluating trigonometric functions: exact values • Cross-Lesson: 11.01 Probability & Statistics

A data set recorded for evaluating trigonometric functions: exact values consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

45 [Challenging] • Target: Evaluating trigonometric functions: exact values • Cross-Lesson: 2.01 Quadratics & Complex Numbers

For the model $f(x) = 5x + 5$ in evaluating trigonometric functions: exact values, evaluate $f(4)$.

a. 25 ✓

b. 30

c. 20

d. 29

Substitute $x = 4$: $f(4) = 5(4) + 5 = 25$.

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